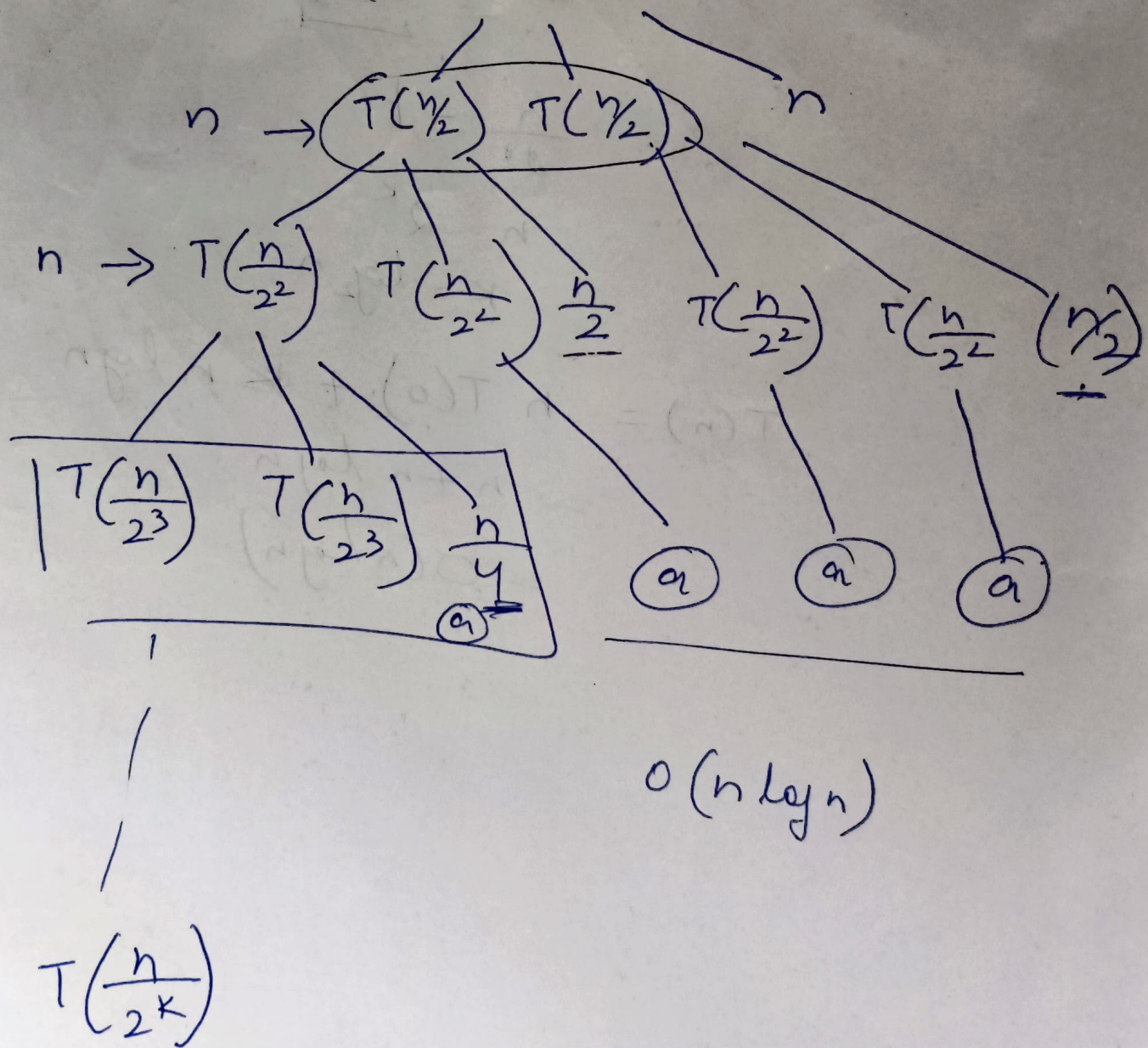
(7)

```
Void Test (int n)
{
  if(n>1)
  {
    for(i=0; i<n; i++)
    {
      statest;
    }
    Test (n/2)
    Test (n/2)
  }
}
```

$$T(n) = 2T(n/2) + n$$



$O(n \log n)$

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

$$T(n) = 2\left(2T\left(\frac{n}{2^2}\right) + \frac{n}{2}\right) + n$$

$$T(n) = 2^2 T\left(\frac{n}{2^2}\right) + 2n$$

$$T(n) = 2^3 T\left(\frac{n}{2^3}\right) + 3n$$

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

$$\frac{n}{2^k} = 1$$

$$n = 2^k$$

$$k = \log_2 n$$

$$T(n) = n\,T(0) + n\log n$$

$$= n + n\log n$$

$$= O(n\log n)$$

Decreasing function Analysis

1. $T(n) = T(n-1) + 1 \rightarrow O(n)$
2. $T(n) = T(n-1) + n \quad O(n^2)$
3. $T(n) = T(n-1) + \log n \quad O(n \log n)$
4. $T(n) = T(n-1) + n^2 \quad O(n^3)$
5. $T(n) = T(n-2) + 1 \rightarrow \frac{n}{2} = O(n)$
6. $T(n) = T(n-100) + n \rightarrow O(n^2)$
7. $T(n) = T(n-500) + n \rightarrow O(n^2)$
8. $T(n) = 2T(n-1) + 1 \rightarrow O(2^n)$
9. $T(n) = 3T(n-1) + 1 \rightarrow O(3^n)$
10. $T(n) = 2T(n-1) + n \rightarrow O(n 2^n)$

MASTER theorem

$$T(n) = a\,T(n/b) + f(n)$$

$a \geq 1$, $b > 1$

$$f(n) = \theta(n^k \log^p n)$$

Case 1: if $\log_b^a > k$ then $\theta(n^{\log_b^a})$

Case 2: if $\log_b^a = k$

- if $p > -1$ $\quad \theta(n^k \log^{p+1} n)$
- if $p = -1$ $\quad \theta(n^k \log\log n)$
- if $p < -1$ $\quad \theta(n^k)$

Case 3: if $\log_b^a < k$

- if $p \geq 0$ $\quad \theta(n^k \log^p n)$
- if $p < 0$ $\quad O(n^k)$

---

① $T(n) = 2\,T(n/2) + 1 \Rightarrow (n^k \log^p n)$ ($k=0$, $p=0$)

$n^0 (\log n)^0 = 1$

$a = 2$, $b = 2$, $k = 0$, $p = 0$

$\log_b^a = \log_2^2 = 1$.

$\theta(n)$

② $T(n) = 4T(n/2) + n$

$a = 4,\ b = 2,\ K = 1,\ P = 0$

$\log_2 4 = 2$

$\Theta(n^2)$

③ $T(n) = 8T(n/2) + n$

$a = 8$, $b = 2$, $K = 1$, $P = 0$

$\log_2 8 = 3$

$\Theta(n^3)$

④ $T(n) = 9T(n/3) + 1$

$a = 9$, $b = 3$, $\log_3 9 = 2$, $K = 0$, $P = 0$

$\Theta(n^2)$

⑤ $T(n) = 4T(n/2) + n^2 \log^2 n$

$a = 4$, $b = 2$, $K = 2$, $P = 2$, $\log_2 4 = 2$

$\Theta(n^2 \log^3 n)$

~~$n^2 \log n$~~

⑥ $T(n) = 2T(n/2) + n^2$

$= a = 2$, $b = 2$, $\log_2^2 = 1$, $K = 2$, $P = 0$, $O(n^2)$

$+ (n^2 \log n)$

$+ n^2$

⑦ $T(n) = 2T(n/2) + \frac{n^2}{\log n}$

$a = 2$, $b = 2$, $K = 2$, $P = -1$

⑧ $T(n/2) + 1$

$a = 1$, $b = 2$, $K = 0$, $P = 0$ $\Rightarrow \log(n)$, $\log_2^1 = 0$

⑨ $T(n) = 2T(n/2) + n$

$a = 2$, $b = 2$, $\log_2^2$, $K = 1$, $P = 0$

⑩ $T(n) = 2T(n/2) + n \log n$ $\rightarrow$

Root function

$T(n) = T(\sqrt{n}) + 1$

$n = 2^m \Rightarrow m = \log n$

$T(2^m) = T(2^{m/2}) + 1$

$S(m) = T(2^m)$

$S(m) = S(m/2) + 1$

1 ... $\log m$

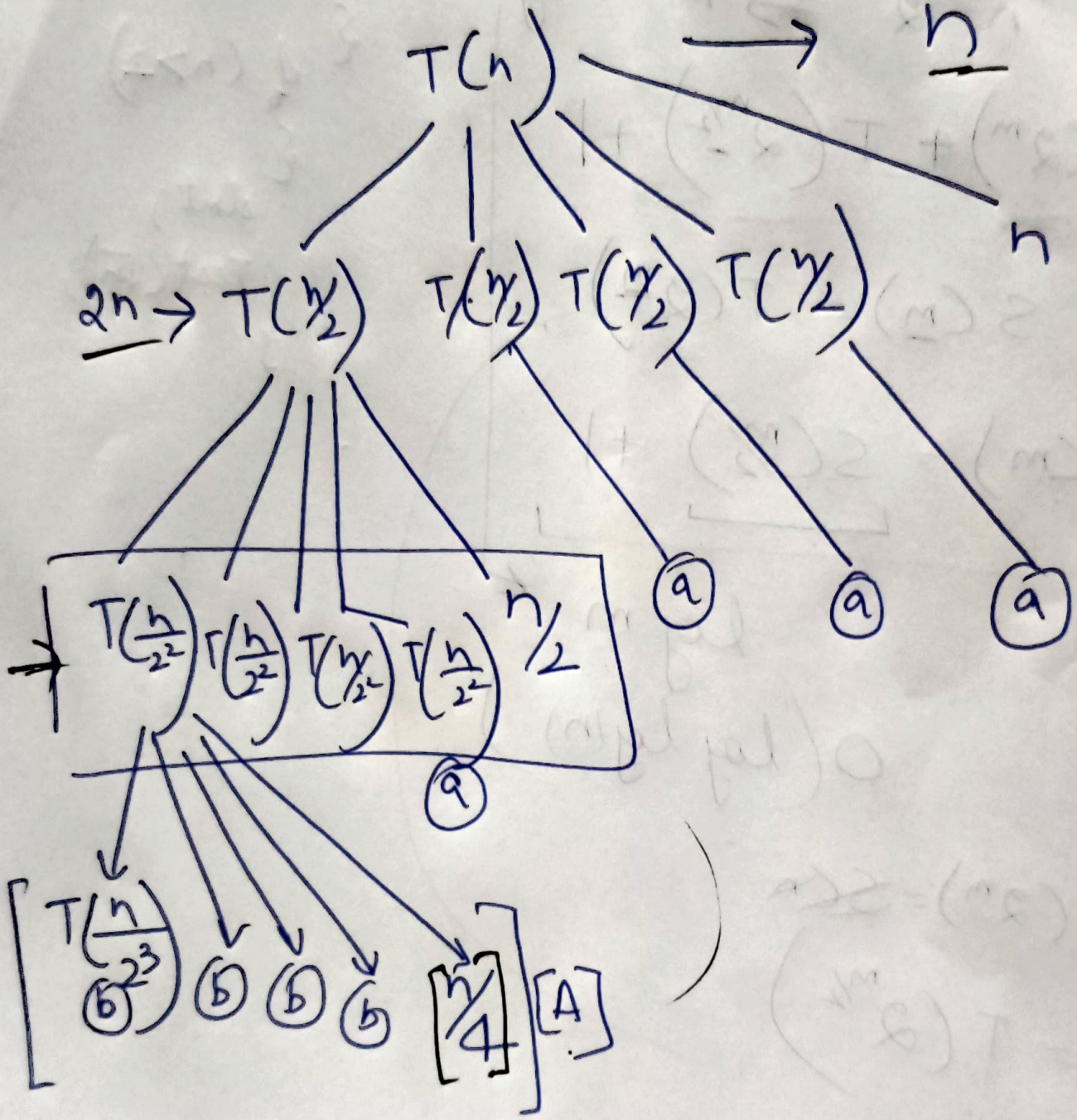
$O(\log \log(n))$

$S(n) = T(2^m) = S(m)$

$S(m/2) = T(2^{m/2})$

```
void Test (int n)
{ if (n>2)
  {
    stmt;
    Test(√n);
  }
}
```

$$T(n) = 4T(n/2) + n$$


T(n) → n
2n → T(n/2)
T(n/2)
T(n/2)
T(n/2)
n
T(n/2²)
T(n/2²)
T(n/2²)
T(n/2²)
n/2
a
a
a
a
T(n/2³)
b
b
b
b
n/4
A